

Metallic nanowires created by biopolymer masking

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(Received 21 April 1999; accepted for publication 1 September 1999)

Nanometer-scale structures are of great interest as potential building blocks for future electronic devices. One significant challenge is the electrical connection of such blocks by nanowires. We present here the use of filamentous biopolymer structures (microtubules) as a template for the creation of metallic nanowires. The microtubules served as a mask in an etching step of a thin metal layer. This approach results in formation of metallic paths (wires) of well-defined dimensions, with a shape determined by the biological template. Electrical measurements confirmed the realization of conductive metallic nanowires. © 1999 American Institute of Physics. [S0003-6951(99)01344-3]

The realization of structures in the lower nanometer range is of great importance for further minimization of electronic devices. High-resolution methods such as electron beam lithography and scanning probe-based surface modification have been developed for nanostructuring application. They are based on a sequential approach, which results in a low throughput. A parallel approach is needed for technological applications. First experiments used self-assembly in synthetic materials for fabrication of structures in the nanometer range, including periodic arrays,¹ semiconducting superlattices,² or a quantum dot network.³

However, the defined wiring of individual elements of such complex systems remains difficult. Molecular units assembled by highly specific interaction, which is a potential of molecular nanotechnology, could provide a wiring mechanism. A needed basic structure is the nanowire, representing a conductive structure with a cross-section in the lower nanometer range, and lengths of several microns. Experiments using biomolecules for nanowire fabrication based on metal deposition at microtubules⁴ or along immobilized DNA molecules⁵ were reported. However, the exact dimensions (and therefore the electrical properties) of such amorphous ablations are hardly controllable. For the creation of nanowires with defined dimensions another approach has to be developed.

One possible technique for nanowiring is a combination of biomolecule templating and planar microstructuring, based on pattern transfer from the biomolecule into a thin conductive layer (Fig. 1). Thereby the thickness of the layer (and therefore one dimension of the created nanowire) is well defined and tunable, and the other dimensions are determined by the shape of the masking molecule.

This letter describes the realization of such an approach using microtubules (MTs). MTs represent rigid rod-like protein assemblages with a well defined diameter (in native state about 24 nm; after flattening about 40 nm,⁶ and lengths of up to several tens of microns). These structures are essentially involved in cell architecture and cellular transport. For adsorption, MTs purified from porcine brain by three cycles of temperature-dependent disassembly/reassembly^{7,8} were sta-

bilized by taxol, which reduces the amount of nonassembled tubulin to negligible values,⁹ and applied to the microelectrode structure described above. After 5 min incubation, the substrate was washed repeatedly with buffer and distilled water containing taxol followed by air drying.

Individual adsorbed MTs were wired using contamination lithography¹⁰ as described elsewhere¹¹ to realize an electrical circuit through the nanowire underneath the adsorbed MTs. Briefly, contamination lines connected prestructured microelectrodes with each end of the MT. The lines were written by repeated exposure of the gold-coated substrate surface to the electron beam of a scanning electron microscope (SEM), and they were used as a mask (similar as the MTs themselves) in the following dry-etching step using Ar ions.

Gold microelectrode structures (thickness of 100 nm) with gap sizes of 12 μm were photolithographically structured onto thermally oxidized silicon wafer after priming with titanium. The whole substrate was sputter coated with 30 nm gold [Fig. 1(a)] prior to biopolymer adsorption [Fig. 1(b)]. Samples without gold coating were used as control.

A scanning force microscope (SFM) Dimension 3000 (Digital Instruments, Santa Barbara, CA) was used for characterization of the MT adsorption. Then, samples were SEM imaged (DSM 960, Zeiss, Germany) without additional gold coating prior to writing contact lines by contamination lithography as described above [Figs. 2(a)–2(d)]. Control samples, which had no initial gold coating, were gold coated (~ 15 nm) before SEM-imaging [Figs. 2(e)–2(h)].

I – U -plots curves were measured using a Keithley Sourcemeter 2400. A voltage range from -3 to 3 V was used. The current was limited to 1 μA to avoid nanowire damages.

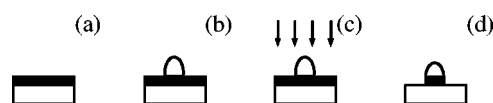


FIG. 1. Principle of biopolymer-masked structuring. Cross-section. (a) The substrate (white; e.g., silicon oxide) is coated with a thin conductive layer (black; e.g., 30 nm gold). (b) A biopolymer is adsorbed onto the coated substrate. (c) An etching step (e.g., dry etching) removes the conductive layer. (d) Only the surface region masked by the biopolymer remains metal-coated resulting in a nanowire.

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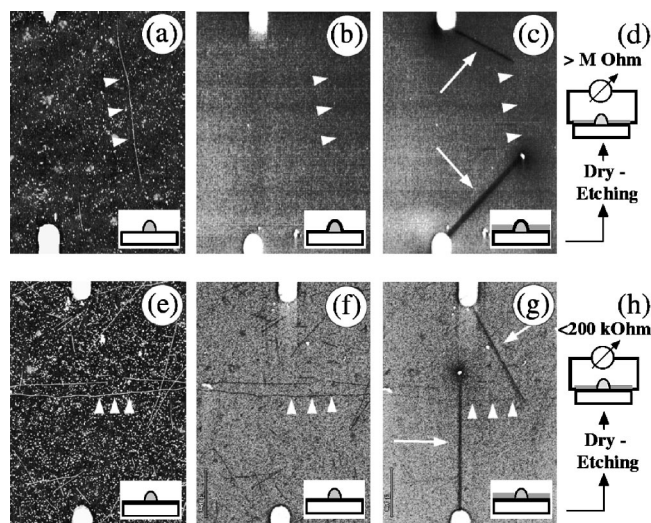


FIG. 2. Biotemplated fabrication of nanowires. The insets illustrate each step of the procedure by showing the substrate (white) with adsorbed MTs (gray), a gold coating (black), and contamination lines (dark gray). Top (a)–(d): Control experiment without gold layer in-between MTs and the substrate. (a) MT (arrowheads) were adsorbed directly to the silicon oxide substrates, which contain prestructured microelectrodes with a gap width of 12 μm , and visualized by SFM. (b) A gold coating was applied on top of the adsorbed MTs, and the same sample region as seen in (a) was visualized by SEM. (c) Two contamination lines (arrows) were written to wire the MT (arrowheads) with the microelectrodes. (d) Electrical measurements after dry etching (to remove the gold coating) yielded a very high resistivity (more than 10 G Ω) of the MT structures. Note that there is no gold directly beneath the MT. Bottom (e)–(h): Use of biopolymers for nanowire fabrication. (e) Scanning force micrograph of MTs (arrowheads) adsorbed onto a gold-coated substrate in the gap of microelectrodes. (f) The same sample imaged by SEM. (g) Contamination lines (arrows) connect the MT with the microelectrodes. (h) After dry etching, electrical measurements demonstrate a resistivity of less than 200 k Ω for a circuit including the gold nanowire beneath the MT.

Scanning force micrographs (Fig. 2 left) of MTs adsorbed onto a silicon oxide (a) or gold (e) substrate yielded the typical elongated structure of MTs (arrowheads). MTs were chosen as molecular template, because their dimensions provide a connection between nanometer structures and the micrometer world of electrodes prepared by conventional microlithography. MTs exhibit a high stiffness compared to other biomolecules like DNA. This advantage is accompanied by a good adsorption to technical substrates (glass, silicon oxide, gold) in the presence of microtubule-associated proteins (MAP). In addition, MAPs were found to increase the flexural rigidity of MTs.¹² The MAP-mediated adsorption is salt dependent and reversible,¹¹ thus opening the way for controlled ad-/desorption by changing the salt conditions.

The increased background roughness visible on the gold substrate [Fig. 2(e)] is due to globular structures with dimensions in the lower nanometer range. Because the roughness of the empty gold surface is more than one order of magnitude lower (data not shown), we assume that the observed features are due to an increased adsorption of assembly-inactive tubulin di- or oligomers. Such an adsorption would be in agreement with the known affinity of proteins for gold surfaces.¹³

MTs adsorbed onto silicon oxide substrates were gold coated [cf. Fig. 2(b)] for SEM imaging. The SEM [(b) and (c)] did not reveal the MTs (arrowheads), which complicates the positioning of contamination lines. To overcome this

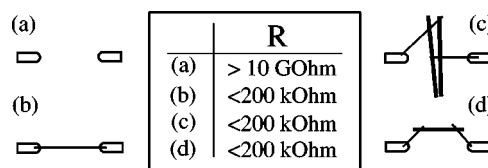


FIG. 3. Conductivity tests demonstrating the creation of a nanowire beneath the adsorbed microtubule. The resistivity of the following samples are given: (a) microelectrodes without connection (negative control), (b) microelectrodes bridged by a gold line structured by contamination lithography (positive control), (c) and (d) two samples of MT-templated gold nanowires (contacted by contamination lines).

problem, SFM images [Fig. 2(a)] were used to get starting point, angle, and length of each contamination line needed. Then, the actual writing of the lines was conducted blindly based on these parameters. Figure 2(c) shows the resulting lines (arrows).

MTs adsorbed onto gold-coated substrates are shown in the lower part of Fig. 2. Beside the increased background, the SFM image (e) is comparable to images of MTs adsorbed to silicon oxide. However, the SEM images of MTs adsorbed to gold allow the localization of MTs as dark shadows [arrowheads in Figs. 2(f) and 2(g)], so that the positioning and writing of contamination lines is straightforward [arrows in Fig. 2(g)].

A dry-etching step transferred the shape of the adsorbed MTs into the gold layer [Fig. 1(d)]. The gold is removed by accelerated argon ions, except at surface parts directly protected by MTs. Timing of the etching step is essential: The gold layer at the unprotected surface area has to be removed, but the parts beneath the MTs should be preserved. An electrical characterization was used for monitoring this step: The gold lines beneath the MTs were wired to microelectrodes, and their conductivity was compared to control measurements using empty microelectrode gaps. Wiring was realized using contamination lithography.¹⁰ Briefly, electron beam-induced deposition of amorphous material created in a SEM is used as a mask of the gold layer in an etching step. Such gold lines with dimensions of 30 nm (thickness) and 50–100 nm (width) exhibit sufficient electrical conductivity, as demonstrated by a positive control (line wiring microelectrode gap directly) and previous measurements (Fig. 3).¹¹

Lines produced on the base of contamination lithography wired two points along MTs to microelectrodes. After etching, such setups yielded a low resistivity (< 200 k Ω), comparable to test structures made by contamination lithography.¹¹ Control measurements using adjacent gaps without molecules resulted in a high resistivity (> 10 G Ω), which excludes any substantial electrical flow over the substrate and/or gold residues. This observation confirms the successful preparation of conductive nanowires using biomolecular templates with diameters in the lower nanometer range.

To exclude any contributions of the MT to the observed conductivity, a control experiment without underlying gold was conducted [Figs. 2(a)–2(d)]. MTs were adsorbed to the substrate [Fig. 2(a)] and wired using contamination lithography, which includes gold coating [Fig. 2(b)], writing contamination lines [arrows in Fig. 2(c)], and removal of the gold by dry etching [Fig. 2(d)]. These steps result in gold

lines connecting the microelectrodes with a MT, so that the resistivity of the MT can be probed. Electrical measurement yields a resistivity higher than $10\text{ G}\Omega$, thereby excluding any significant contribution of the MTs to the observed conductivity of the created nanowires.

Biopolymers provide a great variety of structures and patterns, which might be used as masks. Biomolecular self-assembly could be applied for construction of complex two- and three-dimensional (2D) and 3D structures in the nanometer range, e.g., by using proteins and/or DNA.^{14–16} However, it is difficult to position such structures on defined locations of microstructured surfaces, as necessary for incorporation into conventional microstructured environments. Alignment of elongated molecules in a liquid flow, receding meniscus, or electrical field could be a first step of positioning. Such effects were demonstrated in experiments with a polymer,¹⁷ DNA,^{18–20} and MTs.^{11,21} Another problem is the positioning of the starting point, which could be addressed by specific binding. Using this effect, long DNA molecules were bound to modified gold electrodes.⁵ Techniques based on this approach are promising for the development of hybrid structures including nanostructures wired to (micro-) technical environments.

Summarizing, we demonstrated the use of biopolymer templates for nanowire structuring as a tool for producing nanometer-sized metallic structures. Although wiring of the nanowire to prestructured electrodes could be accomplished, further development is necessary to incorporate nanowires into a microelectronic environment as prerequisite for future applications.

The authors thank M. Sossna and H. Porwol for substrate preparation and ion etching; S. Jakobs and A. Duparé for SFM access; S. Häfner for technical assistance with microtubule preparation; F. Jahn for SEM measurements and writing contamination lines. Funded by the German Ministry

for Education and Research (BMBF) and the Fonds der Chemischen Industrie.

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